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Abstract

FAIRE: Force-Aware Imitation with Residual-Removal-Aware Blending for CAD-Free Robotic Surface Repair

Localized surface defects must be removed selectively because unnecessary processing can damage intact material, whereas insufficient processing leaves the defect visible or functionally significant. Conventional perception-and-planning approaches can localize a defect and generate a geometric repair path, but defect geometry alone does not specify the process intensity, local repetition, desired contact behavior, or termination condition required for removal when depth, severity, and removal difficulty are not explicitly available. FAIRE therefore uses visual and force-conditioned imitation learning as an expert repair policy. From an image, robot state, and recent six-axis wrench history, the policy predicts short-horizon Cartesian pose and desired three-axis force trajectories that represent the operator's defect-dependent decisions about how to process, how much to process, where to repeat, and when to terminate. The desired contact force is a controller-level reference, and the demonstrated tangential feed velocity is preserved during this selective-removal stage.

Successful defect removal, however, does not guarantee a spatially uniform and smoothly restored surface. Concentrated processing can leave a local depression, nonuniform removal, turning-point overlap, or a visible boundary to the untreated surface. FAIRE therefore logs the actual executed pose, orientation, wrench, and velocity history and converts it into a Preston-model-based estimated material-removal map using a calibrated Preston coefficient

and tool influence function. A target profile regularizes the defect-containing core toward a uniform target depth and smoothly decreases removal across an outer transition region. The difference between this profile and the estimated removal map defines the residual-removal demand.

A model-based planner generates a local CAD-free blending path and nominal process conditions from that demand. Admittance control compensates for unknown surface height, while bounded residual reinforcement learning adapts roll, pitch, and feed velocity during blending on unseen smooth curved surfaces. Velocity adaptation is residual-removal-aware and is not applied during expert imitation. Evaluation is planned through staged comparisons of expert imitation, force-history representations, removal-model calibration, blending, residual policies, and end-to-end repair. Quantitative findings will be populated only from completed experiments.

Keywords: Robotic Surface Repair, Force-Aware Imitation Learning, Material-Removal Estimation, Model-Based Blending, Admittance Control, Residual Reinforcement Learning, CAD-Free Surface Processing

Chapter 1

Introduction

1.1 Background and Motivation

Localized surface repair must remove a defect without unnecessarily processing the intact material around it. This requirement is more demanding than visiting a region with a tool. Contact force, true tool–surface relative velocity, orientation, dwell, overlap, and material response jointly determine what is physically removed [1–3]. A robot must therefore decide how a defect should be treated, how much processing it requires, which locations require repetition, and when selective removal should terminate.

Geometric perception and coverage planning provide essential information about where the robot can move [4–6]. They do not, by themselves, reveal defect depth, local severity, removal difficulty, or the appropriate process intensity. These variables may be only partially observable in an RGB image and may become apparent only through interaction. FAIRE addresses this gap with a force-aware imitation policy that learns an operator's defect-conditioned repair decisions from visual context and contact history.

1.2 Localized Defect Removal versus Surface Restoration

Selective removal and surface restoration are related but distinct objectives. The first objective is to eliminate the defect while limiting collateral processing. The second is to leave a spatially regular surface that connects smoothly to its surroundings. An imitation policy may successfully remove a visible scratch

Localized defect
↓ expert IL selective removal
Defect removed, but localized depression or nonuniformity may remain
↓ execution-derived removal estimation and model-based blending
Uniform core and smoothly decaying transition to the intact surface

Figure 1.1: Motivation for separating selective defect removal from complete surface restoration. The final illustrated result sequence will be populated from verified experiments.

while concentrating material removal near the defect. The resulting surface may contain a local depression, nonuniform removal, excessive overlap at turning points, or a height and appearance discontinuity at the processed boundary.

Consequently, defect removal does not imply complete surface restoration. FAIRE assigns uncertain, defect-conditioned selective removal to expert imitation learning (IL). It assigns quantitative spatial regularization after IL to a model-based blending stage. The executed IL process is measured and converted into a spatial removal estimate so that blending responds to what the robot actually produced rather than to the commanded path alone.

1.3 Why Defect Localization Is Not Sufficient

Conventional perception-and-planning approaches can localize a defect and generate a geometric path through its image-space or surface-space region. A region of interest nevertheless does not uniquely determine the necessary force, tangential feed velocity, number of passes, local repetition pattern, or stopping condition. Two defects with similar visible footprints may differ in depth or removal difficulty, while different parts of one defect may demand unequal treatment. A purely geometric recipe must receive these process decisions from another source.

FAIRE does not claim a separate explicit autonomous detector. Within a bounded task workspace, the visual policy may implicitly localize and interpret the defect together with the robot state and recent six-axis wrench history. Its role is summarized as follows: *the imitation policy determines how and to what extent a visually observed defect should be selectively removed from visual context and interaction history.*

1.4 Skill and Expert Roles in Robotic Surface Repair

A *skill* executes a process whose region, target, and parameters have already been specified. An *expert* interprets uncertain observations and determines the appropriate behavior, process intensity, repetition, and termination. This distinction prevents geometric execution from being conflated with defect-dependent decision making.

In FAIRE, force-aware IL acts as the expert repair policy. Model-based blending acts as a structured restoration skill. The latter uses the material-removal distribution produced by IL to regularize the inner repair region toward a target depth and to reduce removal gradually across an outer transition region. It therefore receives a quantitatively defined target instead of deciding the severity of the original defect.

1.5 Research Questions

RQ1: Can force-aware imitation learning infer and reproduce a defect-conditioned expert repair strategy when the required process intensity and removal termination are not explicitly specified?

RQ2: Can the executed pose–wrench–velocity history be converted into an accurate spatial material-removal estimate using a calibrated Preston model and tool influence function?

RQ3: Does residual-removal-based model planning improve the uniformity of the defect-containing core and the smoothness of the transition to the surrounding surface compared with IL-only repair or predefined uniform processing?

RQ4: Do bounded residual orientation and feed-velocity policies improve contact stability and target-removal accuracy during CAD-free blending on previously unseen smooth curved surfaces?

Force-action interfaces and recurrent wrench representations are evaluated as subsystem ablations under RQ1 rather than as thesis-level objectives.

1.6 Contributions

The intended integrated contributions are:

1. a VR-calibrated motion–force teaching and force-aware imitation framework that transfers an operator's defect-conditioned selective repair strategy rather than only reproducing a predefined geometric path;

2. an execution-derived spatial material-removal representation that converts actual pose, wrench, and velocity history into a calibrated Preston-model-based removal map;

3. a residual-removal-aware model-based blending method that regularizes the defect-containing core to a target removal depth and gradually reduces removal across a transition band; and

4. a CAD-free adaptive execution architecture in which admittance control handles unknown surface height and bounded residual RL adapts tool orientation and local feed velocity during blending on unseen smooth surface geometries.

Preston's equation, admittance control, GRUs, coverage planning, and residual RL are not claimed as novel in isolation.

1.7 Scope, Assumptions, and Thesis Status

The scope is local repair on smooth, continuously varying surfaces with moderate convex or concave curvature. It excludes sharp edges, steps, grooves, discontinuities, and extreme curvature unless later verified experimentally. CAD-free means that the planner does not require a complete CAD model or full 3D reconstruction; it does not mean geometry-free. A bounded local workspace, a viable local 2D coverage parameterization, and compliant contact are still assumed.

Hardware identifiers, network dimensions, control parameters, removal-model calibration, RL details, metrology settings, and quantitative results are not yet established in the source repository. They are marked [Planned/TBD: to be determined from the final implementation] or [Planned/TBD: to be populated from experimental results]. No unmeasured performance advantage is asserted.

1.8 Thesis Organization

Chapter 2 reviews the relevant literature. Chapter 3 formulates the problem and allocates module responsibilities. Chapter 4 presents motion–force demonstration collection. Chapter 5 defines expert IL and its compliant execution. Chapter 6 derives the execution-based removal and residual maps.

Chapter 7 presents CAD-free blending with residual RL. Chapter 8 defines the staged evaluation. Chapter 9 concludes the thesis, and Appendix A records reproducibility requirements.

Chapter 2
Related Work

2.1 Robotic Surface Repair and Polishing

Robotic surface processing combines task specification, path planning, contact regulation, and process monitoring. Coverage planners reason about workpiece boundaries, tool footprints, reachability, and collision constraints, while polishing controllers regulate contact along curved surfaces [2, 4, 5]. Integrated surface-treatment systems demonstrate the value of connecting inspection, planning, and controlled execution [6, 7]. FAIRE focuses specifically on the division between uncertain defect-dependent treatment and quantitative post-removal blending.

2.2 Vision-Based Defect Localization and Geometric Repair Planning

Vision can identify candidate defect regions and provide geometry for a local path. Such localization answers where a defect appears, but a repair controller also needs a target removal amount and process conditions. These values are especially uncertain when appearance is an incomplete proxy for depth and material response. FAIRE therefore treats visual context as part of an expert policy observation instead of assuming that an image-space region fully specifies the repair.

2.3 Imitation Learning for Contact-Rich Manipulation

Visuomotor imitation policies learn temporally coherent actions from demonstrations [8, 9]. Contact-rich manipulation adds latent interaction state: similar images and poses can correspond to approach, stable contact, excessive loading, or contact loss. Force-aware methods incorporate wrench observations or force-related actions to reduce this ambiguity [10–13]. FAIRE uses this principle to learn defect-conditioned selective treatment rather than generic trajectory replay.

2.4 Force-Aware Motion–Force Demonstration Learning

External teaching devices can capture human motion independently of a robot embodiment, while force/torque instrumentation records the accompanying interaction. ForceMimic provides a relevant precedent for force–motion capture in contact-rich tasks [11]. FAIRE retains a handheld device equipped with a six-axis force/torque sensor, VR-to-robot calibration, and synchronized image, pose, orientation, and wrench recording. Its subsystem study compares current-wrench and wrench-history observations and MLP, LSTM, and GRU encoders without claiming novelty for recurrence itself.

2.5 Admittance and Compliant Contact Control

Impedance and admittance control specify compliant relations between motion and interaction force [14–16]. Admittance control is appropriate when a policy provides a desired contact force but the robot accepts Cartesian motion references. In FAIRE, desired contact force always denotes a controller-level reference. The controller changes the normal-direction position to track it; the policy output is not a direct actuator force.

2.6 Material-Removal Modeling and Tool Influence Functions

Preston's model relates removal to pressure and relative velocity [1]. Spatial polishing models further require a tool influence function (TIF) describing how contact distributes removal around a tool pose. A usable repair map must combine calibrated process response, normal force, true tool–surface relative velocity, dwell, overlap, and tool geometry. FAIRE introduces this established modeling principle as the bridge from executed IL behavior to a spatial estimate; it does not claim Preston's relation or the TIF concept as new.

2.7 Robotic Blending and Coverage Planning

Coverage planning generates raster, contour, or related paths over a defined region and assigns spacing, overlap, and process conditions [4, 5]. For localized blending, uniform processing with an abrupt boundary is undesirable because it can preserve a step between processed and intact material. FAIRE instead defines a uniform target in a core region and a smoothly decreasing target in a transition band, then plans only the additional removal demanded by their residual map.

2.8 Residual Learning for Surface and Process Adaptation

Residual learning places bounded learned corrections around a nominal controller, limiting the policy's responsibility and preserving an interpretable

Table 2.1: Functional positioning of FAIRE. Entries describe responsibilities rather than claims of algorithmic novelty.

Component	Input or target	Responsibility	Boundary
Expert IL	Image, robot state, six-axis wrench history	Infer defect-conditioned selective treatment	Does not produce a quantitative restoration profile
Removal model	Actual execution history	Estimate spatial processing/removal produced by IL	Proxy until K and TIF are calibrated
Blending planner	Target and residual-removal maps	Specify local coverage and nominal process conditions	Does not infer original defect severity
Admittance	Desired contact force	Adapt normal position to unknown height	Force is a controller-level reference
Residual RL	Local blending state and residual demand	Adapt roll, pitch, and tangential feed velocity	Active only during blending

baseline. Related contact-rich work supports learned reactive or compliance-aware corrections [17, 18]. In FAIRE, residual RL is restricted to the blending stage. Roll and pitch adapt tool orientation, and a feed-velocity residual responds to spatially varying removal demand and contact stability. Neither residual modifies the demonstrated expert IL stroke.

2.9 Positioning of FAIRE

FAIRE is positioned as an integrated system in which IL acts as an expert and model-based blending acts as a structured restoration skill. Its research claim must be evaluated by the staged experiments in Chapter 8; the present draft reports no positive quantitative result.

Chapter 3

Problem Formulation and FAIRE Overview

3.1 Surface-Repair Objective

Let Ω denote a bounded local repair workspace on a surface, and let $q \in \Omega$ denote a point in its local two-dimensional parameterization. The objective is first to remove a visually observed defect selectively and then to obtain a spatial material-removal profile that is uniform around the defect and smoothly connected to the intact surface. The system must avoid excessive force, workspace violation, contact loss, and overprocessing subject to implementation-specific safety limits.

3.2 Skill–Expert Problem Formulation

A geometric skill receives a target region and process parameters and executes them. An expert instead infers appropriate treatment from uncertain evidence. FAIRE assigns the mapping from visual/contact evidence to selective repair behavior to expert IL, while assigning a quantitatively defined residual-removal objective to the blending skill.

3.3 Expert IL for Selective Defect Removal

The conceptual IL observation is

$$o_t^{\mathrm{IL}} = \{I_t, s_t, w_{t-K:t}\}, \tag{3.1}$$

Geometric skill:
predefined target + process parameters → structured execution

Expert IL:
image + robot state + wrench history
→ treatment, process intensity, local repetition, and termination

Figure 3.1: Skill–expert distinction used throughout FAIRE. Termination is a policy responsibility only if the final implementation supports learned continuation or stopping.

where I_t is the visual observation, s_t is the current robot/tool state, and $w_{t-K:t}$ is recent six-axis wrench history. The policy predicts

$$\pi_{\mathrm{IL}}(o_t^{\mathrm{IL}}) \to \{x_{t:t+H}^{\mathrm{ref}}, R_{t:t+H}^{\mathrm{ref}}, f_{t:t+H}^{\mathrm{des}}\}, \tag{3.2}$$

where $f^{\mathrm{des}} \in \mathbb{R}^3$ is the desired three-axis force trajectory unless the final implementation establishes a different output. The demonstrated tangential feed velocity is preserved:

$$v_{\mathrm{cmd}}^{\mathrm{IL}} = v_{\mathrm{IL}}. \tag{3.3}$$

Only a deterministic safety governor may reduce or stop IL motion.

3.4 Execution-Derived Material-Removal Estimation

The system logs actual IL execution as

$$\mathcal{T}_{\mathrm{IL}} = \{p_t, R_t, w_t, v_{\mathrm{feed},t}, \omega_{\mathrm{spindle},t}\}_{t=1}^{T}, \tag{3.4}$$

using available measured or estimated values. Commanded pose, measured robot/TCP pose, and estimated contact-point pose are separate quantities. After wrench compensation and contact-frame transformation, this history is converted into a Preston-model-based estimated material-removal map $\hat{h}_{\mathrm{IL}}(q)$.

Until calibration and metrology validate its units, it is called a removal proxy or relative processing-intensity map.

3.5 Target Profile and Residual-Removal Formulation

The target profile contains a defect-covering core Ω_{core} and an outer transition band $\Omega_{\text{transition}}$:

$$h^*(q) = \begin{cases} h_c, & q \in \Omega_{\text{core}}, \\ h_c w(d(q)), & q \in \Omega_{\text{transition}}, \\ 0, & \text{otherwise}, \end{cases} \tag{3.5}$$

where $w(d)$ decreases smoothly from one to zero. The required additional removal is

$$\Delta h(q) = \max\{0, h^*(q) - \hat{h}_{\text{IL}}(q)\}. \tag{3.6}$$

Because removal is irreversible, the nonnegative residual cannot compensate for IL over-removal.

3.6 CAD-Free Adaptive Blending

The residual map determines where additional processing is required and provides a quantitative demand for a local raster or contour planner. The planner selects nominal force, tangential feed velocity, overlap, path spacing, and pass count. Admittance control adjusts normal position for unknown height, and residual RL adapts orientation and feed velocity on smooth unseen geometry.

VR-calibrated handheld teaching
↓ synchronized image–pose–orientation–wrench data
Expert IL selective repair
↓ actual execution logging
Preston-model-based removal map
↓ target core/transition profile and residual map
Model-based local blending plan
↓ admittance height adaptation
+ residual roll/pitch and feed-velocity adaptation

Figure 3.2: End-to-end FAIRE architecture. Residual orientation and velocity policies operate during model-based blending only.

3.7 Module Responsibilities

3.8 Assumptions and Safety Boundary

The framework assumes a local parameterizable region, smooth continuous surface variation, moderate curvature, a compatible compliant tool, and calibrated process conditions. It does not assume online access to a CAD-derived perfect normal. Any surface normal used as privileged training reward information must remain outside the online observation unless actually measured. Deterministic force, velocity, orientation, workspace, and cumulative-process limits bound both stages. Exact limits are [Planned/TBD: to be determined from the final implementation].

Table 3.1: Responsibility allocation in the end-to-end system.

Module	Determines	Does not determine
Expert IL	Defect-dependent motion, force, repetition, and implemented stopping behavior	Post-repair spatial regularization
Removal estimator	Spatial effect of the executed process	New material addition or defect semantics
Target-profile rule	Desired core depth and transition shape	Robot contact response
Blending planner	Coverage geometry and nominal process conditions	Original defect severity
Admittance	Normal-direction height correction	Tangential coverage direction
Residual RL	Bounded blending-only roll, pitch, and feed-velocity corrections	IL execution or global path topology

Chapter 4

VR-Calibrated Motion–Force Demonstration Collection

4.1 Teaching-System Purpose

The teaching system records how an operator interprets and treats a localized defect, including the motion pattern, tangential feed velocity, desired contact behavior, and local repetition. FAIRE retains a handheld teaching device equipped with a six-axis force/torque sensor and tracked in a virtual-reality (VR) coordinate frame. An RGB camera supplies visual task context. Image, position, orientation, and wrench streams are synchronized so that each action can be associated with the corresponding defect appearance and interaction history.

4.2 Coordinate Frames and Pose Transfer

Let B, V, T, S, E, and C denote the robot base, VR tracking, tracker, force/torque sensor, TCP/tool, and contact-point frames. The demonstrated TCP pose in the robot base is

$$ {}^{B}T_{E}^{\mathrm{demo}}(t) = {}^{B}T_{V}\, {}^{V}T_{T}(t)\, {}^{T}T_{E}. \tag{4.1} $$

The transform ${}^{B}T_{V}$ is obtained from VR-to-robot calibration, ${}^{V}T_{T}(t)$ is tracked over time, and ${}^{T}T_{E}$ is the fixed tracker-to-tool transform. Calibration observations, estimation method, and validation residuals are [Planned/TBD: to be determined from the final implementation].

Robot base $B \leftarrow$ VR frame $V \leftarrow$ tracker T
$\downarrow$ tracker-to-tool calibration
TCP/tool frame $E \rightarrow$ contact-point frame C
synchronized $I_t,\ p_t,\ R_t,\ w_t$, and timestamps

Figure 4.1: Coordinate chain for VR-to-robot demonstration transfer and contact-point representation. Numerical calibration results and the final apparatus drawing remain TBD.

4.3 Wrench Processing and Contact-Point Representation

The sensor produces a six-axis wrench

$$w_t^S = [F_x, F_y, F_z, \tau_x, \tau_y, \tau_z]_t^S. \tag{4.2}$$

The usable wrench requires bias removal, gravity and tool-load compensation, filtering, and transformation to a declared contact or tool frame. When the reference point changes, the spatial wrench transform must account for the associated moment arm; rotating force components alone is insufficient. The same sign convention and reference point must be used in demonstrations, policy observations, force control, removal estimation, and evaluation.

The contact-point pose differs from the measured TCP pose by the TCP-to-contact transform:

$${}^{B}T_C(t) = {}^{B}T_E(t)\, {}^{E}T_C. \tag{4.3}$$

The offset ${}^{E}T_C$ depends on pad/tool geometry and may vary with compliant deformation. The final implementation must state whether the contact pose is measured, estimated from a rigid offset, or corrected by a compliance model. Unverified deformation compensation is not assumed.

4.4 Synchronization and Dataset Construction

An episode is stored conceptually as

$$D_i = \{I_t, p_t, R_t, w_t, v_{\text{feed},t}, m_t\}_{t=1}^{T_i}, \tag{4.4}$$

where m_t denotes available metadata or event labels. Each stream requires timestamps. The synchronization method must report acquisition rates, clock convention, interpolation or resampling, maximum accepted skew, missing-sample handling, and any measured delay. These values are [Planned/TBD: TBD]. Misalignment is particularly harmful because it pairs visual and pose context with a wrench produced at another contact phase.

4.5 Episode Segmentation and Training Windows

Episodes should separate approach, selective processing, local repetition, and release when these phases can be identified reliably. A training sample at time t contains the current image and robot state, the wrench history $w_{t-K:t}$, and a future reference horizon through $t + H$. Boundary handling, padding masks, stride, K, and H are [Planned/TBD: to be determined from the final implementation].

Dataset partitions are made at the episode, defect, or specimen-trial level. Adjacent windows from one episode must not be split across training and evaluation because that would leak nearly identical temporal context. Normalization statistics and augmentation are fitted using training data only.

4.6 Demonstration Quality, Metadata, and Safety

Episodes are rejected or labeled if they contain tracker loss, sensor saturation, timestamp discontinuity, accidental collision, unsafe load, or missing images. Transformed trajectories must pass reachability, collision, workspace, velocity, and force checks before robot execution. Calibration establishes coordinate correspondence but does not guarantee safe motion.

Required records include operator and specimen identifiers, defect preparation, material and coating, tool and abrasive state, spindle setting if applicable, sensor zeroing, calibration version, contact threshold, start pose, processing annotations, and safety events. Appendix A provides the completion checklist.

Chapter 5

Force-Aware Expert Imitation for Selective Defect Removal

5.1 Expert Repair Objective

The expert IL module determines how and to what extent a visually observed defect should be selectively removed from visual context and interaction history. It learns the operator's defect-conditioned local motion pattern, desired contact force, demonstrated tangential feed-velocity profile, and repeated visits. It is not described as an explicit detector, and it is not reduced to replaying a path whose treatment parameters are already known.

5.2 Visual, Robot-State, and Wrench-History Observation

The policy observation follows Eq. (3.1). Separate encoders compute

$$z_t^I = f_I(I_t), \qquad z_t^s = f_s(s_t), \qquad z_t^w = \mathrm{GRU}(w_{t-K:t}), \tag{5.1}$$

and a fusion module produces $z_t = \mathrm{Fusion}(z_t^I, z_t^s, z_t^w)$. The force branch uses six-axis wrench history as a policy observation. Tangential forces and moments can distinguish partial contact, tool tilt, and unstable interaction that a scalar normal force may conceal. The image encoder, robot-state fields, dimensions, GRU hidden size, and fusion design are [Planned/TBD: TBD].

$$
\begin{gathered}
I_t \to \text{visual encoder} \to z_t^I \\
s_t \to \text{robot-state encoder} \to z_t^s \\
w_{t-K:t} \to \text{GRU wrench-history encoder} \to z_t^w \\
\text{fused representation} \to \{x^{\text{ref}}_{t:t+H}, R^{\text{ref}}_{t:t+H}, f^{\text{des}}_{t:t+H}\}
\end{gathered}
$$

Figure 5.1: Force-aware expert imitation architecture. The desired three-axis force trajectory is a controller-level reference; exact network dimensions remain TBD.

5.3 Motion–Force Trajectory Prediction

The short-horizon prediction is

$$\pi_{\text{IL}}(z_t) \to \left\{x^{\text{ref}}_{t:t+H}, R^{\text{ref}}_{t:t+H}, f^{\text{des}}_{t:t+H}\right\}. \tag{5.2}$$

Position and orientation describe demonstrated motion. The desired three-axis force trajectory describes the contact reference to be realized compliantly. Overlapping-horizon selection, temporal ensembling, prediction frequency, and orientation parameterization are [Planned/TBD: to be determined from the final implementation].

5.4 Supervised Learning Objective

A conceptual multitask objective is

$$\mathcal{L}_{\text{IL}} = \lambda_x \mathcal{L}_x + \lambda_R \mathcal{L}_R + \lambda_f \mathcal{L}_f, \tag{5.3}$$

where the terms penalize position, rotation, and desired-force trajectory error. A rotation-aware loss is required for the selected orientation representation. Loss definitions, weights, normalization, padding masks, optimizer, schedule, batch size, training duration, seeds, and checkpoint selection remain

[Planned/TBD: TBD]. Equation (5.3) does not imply a finalized network or training recipe.

5.5 Why Force History Is Required

An instantaneous wrench may be similar during contact establishment, stable processing, or recovery from overload. A history window provides contact trend, persistence, and phase information. The GRU is separated from the visual and robot-state encoders so that current-wrench, flattened-history, LSTM, and GRU variants can be compared while retaining the same policy output and controller. These comparisons test whether temporal contact context supports expert imitation; they do not claim that the GRU is novel.

5.6 Why Geometric Localization Alone Is Insufficient

A localized region describes where processing is allowed or likely needed, but not the correct intensity at each point. Similar image-space defects may demand different force profiles or repetition, and the interaction history may reveal whether a local pass is effective. The expert policy therefore learns a conditional treatment rather than receiving a complete geometric recipe. Experiment I in Chapter 8 tests this necessity against fixed geometric/ROI processing and vision-only variants.

5.7 Admittance-Based Desired-Force Execution

The desired contact force is tracked through admittance dynamics:

$$M_d \Delta \ddot{p}_t + D_d \Delta \dot{p}_t + K_d \Delta p_t = f_t^{\text{meas}} - f_t^{\text{des}}, \tag{5.4}$$

with command composition

$$p_t^{\mathrm{cmd}} = p_t^{\mathrm{ref}} + \Delta p_t^{\mathrm{adm}}. \tag{5.5}$$

The selected controlled axes and sign convention must be stated. The discrete update, virtual parameters, filtering, saturation, contact-state reset, and robot interface are [Planned/TBD: TBD]. The learned force is never interpreted as a direct actuator command.

5.8 Preservation of Demonstrated Feed Velocity

During expert IL, the commanded tangential feed velocity equals the demonstrated IL velocity as stated in Eq. (3.3). A residual velocity policy is not active because it would alter the operator strategy whose defect-dependent behavior is being reproduced. A deterministic governor may only reduce or stop motion when excessive force, contact loss, workspace violation, command discontinuity, or another declared safety condition is detected.

5.9 Local Repetition and Termination

Repeated processing can appear implicitly in the predicted pose sequence or be represented by an explicit continuation decision. The current repository does not establish a learned stopping output. Accordingly, continuation and termination are [Planned/TBD: to be determined from the final implementation]. If execution uses a fixed horizon, operator stop, visual threshold, or explicit rule, that mechanism must be reported and evaluated without being described as learned termination.

5.10 Online Execution and Safety Governor

At each update, the system preprocesses the image, robot state, and compensated wrench; updates the six-axis history; predicts the reference horizon; selects the current pose and desired-force references; integrates the admittance correction; and applies deterministic safety bounds. The actual measured TCP pose, estimated contact pose, compensated wrench, tangential feed velocity, spindle speed if available, commanded references, contact state, and safety events are logged for Chapter 6. Residual RL remains disabled throughout this stage.

Chapter 6

Execution-Derived Material-Removal Estimation

6.1 Process-Data Logging during IL Execution

Removal estimation uses the process that the robot actually executed, not the policy command alone. For every sample, the log distinguishes commanded TCP pose, measured robot/TCP pose, and estimated contact-point pose. It also records the raw and processed wrench, measured tangential feed velocity, spindle angular speed when available, contact state, timestamps, and safety interventions. The execution record is $\mathcal{T}_{\mathrm{IL}}$ in Eq. (3.4).

Commanded pose is retained for controller analysis but is not substituted for contact location when following error or compliance is significant. The mapping from TCP to contact point uses the calibrated offset and pad/tool geometry from Eq. (4.3). Any compliance-related contact-location correction is [Planned/TBD: TBD].

6.2 Contact Frame and Normal-Force Computation

Let f_t be the compensated force expressed in a common frame, and let n_t be the unit tool or estimated surface normal used by the removal model. Normal force is the projection

$$F_n(t) = f_t^{\mathsf{T}} n_t, \tag{6.1}$$

with sign chosen so that compressive contact is positive. Sensor F_z is not assumed to be the normal force unless the sensor frame is verified to be aligned with n_t.

The online system does not assume access to a perfect CAD normal. During IL map construction, n_t may be obtained from the measured tool orientation under an aligned-tool assumption, from a verified local estimator, or from another implemented source. The chosen method and its uncertainty are [Planned/TBD: to be determined from the final implementation]. Samples below a validated contact threshold are excluded from material removal.

6.3 Preston-Based Spatial Removal Model

A discrete spatial model is

$$\hat{h}_{\mathrm{IL}}(q) = \sum_{t=1}^{T} K\, F_n(t)\, G(q; p_t, R_t)\, v_{\mathrm{rel}}(q,t)\, \Delta t, \tag{6.2}$$

where K is the calibrated Preston coefficient, G is a normalized TIF centered and oriented by the contact pose, and $v_{\mathrm{rel}}(q,t)$ is the magnitude of true tool–surface relative velocity at q. The model accumulates overlapping dwell and repeated passes.

Tangential robot feed velocity and true relative sliding are distinct. If the pad rotates, spindle motion may dominate v_{rel}, while feed changes dwell and the path of the influence footprint. The final implementation must calculate v_{rel} from the available kinematics and tool geometry rather than replacing it automatically with v_{feed}.

6.4 Tool Influence Function

The TIF $G(q; p_t, R_t)$ describes the normalized spatial distribution of removal around a tool pose:

$$G(q; p_t, R_t) \geq 0, \qquad \int_{\Omega} G(q; p_t, R_t)\, dq = 1 \tag{6.3}$$

over a sufficiently large local domain. Its shape may depend on pad geometry, compliance, force, tilt, spindle speed, abrasive state, and material. FAIRE will use the simplest calibrated parameterization supported by measurements; a Gaussian, measured lookup map, or another shape is not assumed before calibration.

6.5 Calibration and Metrology

Controlled polishing tests will vary one declared factor at a time over safe operating ranges and measure the resulting removal. Calibration must cover force dependence, relative-velocity or dwell dependence, spatial TIF shape, and repeated-pass accumulation. The Preston coefficient K is fitted only after defining units and the specific tool–abrasive–material condition.

Removal maps will be measured by a verified method such as profilometry, confocal measurement, laser measurement, or another metrology system whose resolution and registration are documented. [Planned/TBD: Select the metrology instrument, calibration specimens, factor levels, fitting procedure, and validation split.] The validation compares predicted and measured depth maps on held-out process conditions or specimens. Until this is complete, $\hat{h}$ is reported only as a removal proxy or relative processing-intensity map.

actual IL contact poses + compensated normal force
+ true relative velocity + calibrated K, G
$\downarrow$ spatial accumulation
$\hat{h}_{\mathrm{IL}}(q)$ compared with $h^*(q)$
$\downarrow \Delta h(q) = \max(0, h^* - \hat{h}_{\mathrm{IL}})$

Figure 6.1: Conceptual construction of the execution-derived removal map and residual-removal map. Experimental maps will replace the placeholder after calibration and metrology.

6.6 Estimated IL Removal Map

The logged trajectory is rasterized in the local repair coordinates, and each valid contact sample contributes the shifted and oriented TIF weighted by normal force, relative velocity, and sample duration. Registration between robot coordinates and the measured surface map must be reported. Map resolution should be fine enough to represent the TIF without implying precision beyond the metrology system.

6.7 Target Core and Transition Profiles

The target profile is defined by Eq. (3.5). The core contains the defect and is regularized toward a constant h_c. The transition band connects this plateau to zero additional removal on the intact surface. A smooth weight can satisfy

$$w(0) = 1, \quad w(D) = 0, \quad w'(0) = w'(D) = 0, \quad 0 \leq w(d) \leq 1, \tag{6.4}$$

where $d \in [0, D]$ measures outward distance through a band of width D. The final taper function and definition of distance are [Planned/TBD: TBD].

The selection of h_c must be explicit and conservative. Candidate rules include a measured defect-depth estimate, a robust percentile or plateau of

$$\underbrace{\overline{\qquad\qquad\qquad}}_{\Omega_{\text{core}}:\ h_c} \searrow \underbrace{\text{smooth decay to } 0}_{\Omega_{\text{transition}}}$$

uniformize the inner repair region → blend gradually into the intact surface

Figure 6.2: Target removal profile with a uniform core and a smoothly decaying transition band.

the IL removal map, or a process-specific target under a coating-thickness safety bound. A noisy maximum is not used without statistical and physical justification. The final rule is [Planned/TBD: TBD].

6.8 Residual-Removal Map

The residual map in Eq. (3.6) assigns minimal additional processing where IL has already reached the target and additional removal where it has not. It also adds the controlled transition processing needed to connect the repaired core to the intact surface. An uncertainty margin or conservative dead band may be required to prevent model error from causing over-removal; its definition is [Planned/TBD: TBD].

6.9 Model Limitations and Uncertainty

Preston behavior may deviate from linearity because of pad wear, abrasive loading, temperature, saturation, compliance, changing contact area, material heterogeneity, and tool tilt. A fixed TIF approximates a process that may vary with force and orientation. Pose, normal, clock, and metrology registration errors propagate into the spatial map. Most importantly, the model can request additional removal but cannot add material after IL over-removes. These limitations motivate conservative targets, held-out validation, uncertainty reporting, and eventual post-pass inspection.

Chapter 7

CAD-Free Model-Based Blending with Residual RL

7.1 Blending Objective

The blending stage converts the residual-removal map into structured additional processing. Its objective is to reduce residual error in the core, follow the decaying transition target, preserve intact surroundings, and maintain stable contact. It is a model-based restoration skill: the spatial target and process demand have already been defined by Chapter 6.

7.2 Local Coverage-Path Generation

The planner generates a local two-dimensional raster or contour path over cells where $\Delta h(q)$ exceeds a declared threshold. It does not require a complete CAD model or full 3D reconstruction. The local parameterization provides coverage direction and spatial indexing; unknown height is handled by admittance, and moderate orientation variation is handled by residual RL.

Path topology, boundary offset, turning strategy, and collision/reachability checks are [Planned/TBD: TBD]. The planned scope is limited to local, smooth, continuously varying surfaces without sharp edges, steps, grooves, or discontinuities.

7.3 Nominal Force, Velocity, Overlap, and Pass Planning

Using the calibrated process model, the planner assigns nominal desired contact force $f_{\text{nom}}^{\text{des}}$, tangential feed velocity $v_{\text{nom}}(q)$, path spacing, overlap, and pass count. These quantities should be chosen together because they jointly determine dwell and accumulated removal. A conservative optimizer or rule-based inversion of Eq. (6.2) may be used; the implemented method is [Planned/TBD: TBD].

7.4 Admittance-Based Surface-Height Adaptation

The planner produces nominal Cartesian position p_{nom}. Admittance control changes its normal component:

$$p_{\text{cmd}} = p_{\text{nom}} + \Delta p_{\text{adm}}, \tag{7.1}$$

so that unknown surface-height variation is accommodated while tracking the desired contact force. The desired contact force remains a controller-level reference. Tangential coverage direction is not generated by admittance.

7.5 Residual Tool-Orientation Adaptation

During blending only, bounded roll and pitch corrections adapt the tool to smooth unseen curvature:

$$R_{\text{cmd}} = R_{\text{nom}} \, \text{Exp}([\Delta\phi_t, \Delta\theta_t, 0]_\times) \,. \tag{7.2}$$

Yaw and coverage direction remain the planner's responsibility unless the final implementation establishes otherwise. The policy must not assume online access to a CAD-derived or perfect surface normal. A simulated normal may be used as privileged reward information during training only if that separation is stated.

7.6 Residual-Removal-Aware Feed-Velocity Adaptation

The velocity command is

$$v_{\text{cmd}} = \text{clip}\left(v_{\text{nom}}(q_t) + \Delta v_t, v_{\min}, v_{\max}\right). \tag{7.3}$$

The objective is to achieve spatially varying target removal at the desired locations. The policy should slow where residual demand is large, move faster where little removal remains, reduce unnecessary processing near target completion, and slow when force, contact, or orientation becomes unstable. Feed velocity is not called true sliding velocity; the latter is reserved for actual tool–surface relative motion in the removal model.

7.7 Residual Policy Observation, Action, and Reward

The residual action is

$$\Delta a_t^{\text{RL}} = [\Delta\phi_t, \Delta\theta_t, \Delta v_t]^{\mathsf{T}}. \tag{7.4}$$

A conceptual observation may include local residual-removal demand, six-axis wrench, force-tracking error, current orientation, current and nominal feed velocity, and local path state:

$$o_t^{\text{RL}} = [\Delta h_{\text{local}}, F_x, F_y, F_z, \tau_x, \tau_y, \tau_z, e_F, R_t, v_t, v_{\text{nom}}, \xi_t]. \tag{7.5}$$

This is not an implemented state specification. [Planned/TBD: Finalize the RL observation, normalization, state estimator, algorithm, network, and update rate.]

local 2D nominal coverage path
$+ \Delta p_{\text{adm}}$ for surface height
$+ [\Delta\phi, \Delta\theta]$ for smooth orientation variation
$+ \Delta v$ for residual-removal-aware tangential feed velocity
$\downarrow$ final bounded Cartesian command

Figure 7.1: Blending command composition. The residual policies modify model-based blending and never correct the demonstrated IL trajectory.

A conceptual reward may penalize residual-removal error, force-tracking error, tangential wrench, action magnitude, action change, contact loss, and safety violation:

$$r_t = -c_h e_h - c_F e_F^2 - c_w \|w_t^{\tan}\|^2 - c_a \|\Delta a_t^{\mathrm{RL}}\|^2 - c_s - c_{\text{safety}}. \tag{7.6}$$

Terms, weights, temporal credit assignment, privileged signals, and termination are [Planned/TBD: TBD]; Eq. (7.6) is only a design envelope.

7.8 Interaction between Orientation and Velocity Policies

Orientation changes contact distribution and measured wrench, while velocity changes dwell, force transients, and removal accumulation. Their coupled effects require a common observation convention, compatible update rates, and final arbitration by safety limits. Experiment V compares nominal execution, classical moment-based orientation adaptation, orientation-only residual RL, and the full orientation–velocity residual. Separate and joint policies remain [Planned/TBD: to be determined from the final implementation].

7.9 Iterative Removal-Map Update and Replanning

After blending pass k, the estimated increment can update the accumulated map:

$$\hat{h}_{k+1}(q) = \hat{h}_k(q) + \Delta\hat{h}_{\text{blend},k}(q), \tag{7.7}$$

$$\Delta h_{k+1}(q) = \max\{0, h^*(q) - \hat{h}_{k+1}(q)\}. \tag{7.8}$$

This defines the loop *Plan* $\rightarrow$ *Execute* $\rightarrow$ *Update Removal Map* $\rightarrow$ *Replan*. [Planned/TBD: Iterative replanning is a planned extension unless verified in the final implementation]; it must not be reported as completed before that verification.

7.10 Safety Limits and Expected Failure Modes

Actions are clipped before command composition. Deterministic limits cover normal and tangential force, moments, feed velocity, orientation correction, correction rate, admittance displacement, workspace, contact loss, cumulative predicted removal, and emergency stop. Expected failures include TIF mismatch, incorrect residual demand, policy saturation, oscillatory orientation or velocity, contact loss, conflict between residual channels, and extrapolation beyond trained curvature. Interventions and saturation frequency are reported as outcomes rather than silently discarded.

Chapter 8

Experimental Evaluation

8.1 Research Questions and Evaluation Strategy

The evaluation follows RQ1–RQ4 from Chapter 1 through six staged experiments. Each stage isolates one responsibility before the end-to-end comparison: the need for expert treatment, force-aware representation, removal-model validity, blending contribution, residual adaptation on unseen geometry, and complete repair progression. All numerical fields remain blank until supported by traceable logs and metrology.

8.2 Robot, Tool, Sensor, Camera, and Specimens

The experimental system requires a robot with a Cartesian command interface, a polishing or repair tool, a six-axis force/torque sensor, a visual sensor, and specimens containing reproducible localized defects. The robot, tool, sensor, camera, spindle, abrasive, material, coating, and metrology hardware are [Planned/TBD: to be determined from the final implementation].

Specimens should cover smooth flat, convex, and concave geometries within a declared curvature range. Sharp edges and discontinuities are excluded. Defect variations may include severity, depth, width, local nonuniformity, required repetition, or removal difficulty only when those conditions can be produced and measured reliably.

8.3 Dataset and Train/Validation/Test Splits

Demonstration windows are split by episode, defect, or specimen rather than by adjacent timestamps. Model-selection trials are separated from final tests. For geometry generalization, RL training surfaces may include flat and selected convex/concave radii, while test surfaces use previously unseen smooth curvature values or combinations. Material, tool, abrasive, and process conditions remain fixed during the geometry study unless cross-material generalization is explicitly evaluated.

Dataset size, split ratios, defect distribution, surface assignments, and seed count are [Planned/TBD: TBD]. The split manifest and preprocessing statistics will be stored with the reproducibility records.

8.4 Experiment I: Necessity of Expert Imitation

Experiment I tests whether localization plus a predefined skill determines sufficient treatment. The planned comparison includes a geometric/ROI path with a fixed motion–force recipe, vision-only IL, vision–force IL, and full force-history IL. Conditions should include visually similar regions that require measurably different process intensity, repetition, or stopping behavior.

Metrics include defect-removal ratio, task success, collateral removal outside the defect, force-tracking error, pass or repetition count, termination error when defined, and task time. The defect-removal and termination metrics require explicit pre/post measurements and reference criteria before use.

Table 8.1: Planned expert IL baseline comparison. Blank entries await experimental results.

Method	Defect removal	Collateral removal	Repetition/ termination	Task time
Geometric/ROI fixed recipe				
Vision-only IL				
Vision–force IL				
Full wrench-history IL				

8.5 Experiment II: Force-Aware IL Representation

Experiment II provides subsystem evidence for RQ1. Action-interface comparisons include pose only, force observation without desired-force output, fixed-force admittance, phase-based force, one-step desired force, and horizon desired-force trajectory. Observation/encoder comparisons include current wrench versus wrench history and MLP, LSTM, and GRU variants. All methods in each subcomparison retain the same data split, policy backbone where applicable, controller, and IL-stage velocity rule.

Table 8.2: Planned force-aware IL ablations. Blank entries await experimental results.

Variant	Force observation	Policy force output	Force RMSE	Contact loss	Defect removal	Collateral removal	Success
Pose only	None	None					
Force observation, pose output	Current/history	None					
Fixed-force admittance	History	Fixed reference					
Phase-based force	History	Phase reference					
One-step desired force	History	One step					
Horizon desired-force trajectory	History	Short horizon					
Current-wrench MLP	Current	Short horizon					
Wrench-history MLP	History	Short horizon					
Wrench-history LSTM	History	Short horizon					
Wrench-history GRU	History	Short horizon					

Table 8.3: Planned removal-model calibration and validation results.

Condition	Depth RMSE	Correlation	Peak error	Mean error	Repeat-ability
Force dependence					
Velocity/dwell dependence					
Repeated-pass accumulation					
Held-out map validation					

8.6 Experiment III: Removal-Model Calibration and Validation

Controlled tests calibrate K and G, verify force and velocity/dwell dependence, and evaluate repeated-pass accumulation. Predicted maps are registered to measured maps obtained with the selected metrology system. Training and validation process conditions or specimens are separated.

Depth RMSE is defined after registration as

$$\mathrm{RMSE}_h = \sqrt{\frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \big(\hat{h}(q) - h_{\mathrm{meas}}(q)\big)^2}, \tag{8.1}$$

where $\mathcal{Q}$ is the declared valid measurement set. Spatial correlation, peak and mean removal error, profile error, map similarity, and repeatability are included only after their registration, masking, and aggregation procedures are specified.

8.7 Experiment IV: Residual-Removal-Based Blending

Experiment IV compares IL only, IL followed by fixed uniform surrounding processing, and IL followed by model-based residual-removal blending. It evaluates whether the proposed target regularizes the core and produces a smooth transition without unnecessary overprocessing.

Core uniformity is reported using a stated dispersion measure over Ω_{core}. Residual-removal RMSE compares achieved and target profiles. Transition-profile error is measured only inside $\Omega_{\mathrm{transition}}$, and boundary slope or gradient

Table 8.4: Planned comparison of IL-only repair and blending strategies.

Method	Core uniformity	Residual RMSE	Transition error	Over-processing	Appearance
IL only					
IL + uniform surrounding process					
IL + residual-removal blending					

is computed using a declared spatial derivative and sampling method. Surface roughness, gloss, haze, or another appearance metric is reported only if a verified instrument or controlled protocol is available.

8.8 Experiment V: Residual RL on Unseen Surface Geometries

The residual study compares fixed orientation with nominal model velocity, classical moment-based orientation adaptation with nominal velocity, orientation-only residual RL, and full residual orientation–velocity RL. Training and test geometries are disjoint. A classical moment-based method is included only after its gains, frames, and saturation rules are specified.

Normal-force RMSE uses the projected force from Eq. (6.1). Other metrics include contact-loss rate, tangential-force magnitude, τ_x/τ_y, orientation smoothness, residual-removal error, core uniformity, transition error, task time, safety interventions, and policy saturation frequency.

Table 8.5: Planned residual RL ablation on unseen smooth surface geometries.

Method	Force RMSE	Contact loss	Tangential force	Orientation smoothness	Removal error	Task time	Interventions	Saturation
Fixed orientation + nominal velocity								
Moment-based orientation + nominal velocity								
Orientation-only residual RL								
Residual orientation + velocity RL								

Table 8.6: Planned end-to-end results at each repair stage.

Stage	Defect measure	Core uniformity	Transition error	Collateral removal	Appearance
Before repair					
After expert IL					
After model-based blending					

before repair | after expert IL | after blending

registered appearance, surface profile, and removal-map panels
[Planned/TBD: to be populated from experimental results]

Figure 8.1: Planned end-to-end visualization of selective removal and subsequent spatial regularization.

8.9 Experiment VI: End-to-End Surface Repair

End-to-end evaluation records the surface before repair, after IL selective removal, and after model-based blending. The intended analysis asks whether IL removes the defect selectively, whether IL-only removal remains spatially nonuniform, and whether blending brings the core and transition closer to the declared target. These are hypotheses, not current findings.

8.10 Metrics

Every metric will state its physical meaning, units, instrument or data source, valid spatial/temporal mask, registration method, aggregation, and treatment of failed trials. Force-tracking error uses desired and projected measured force in a common frame. Contact loss uses a calibrated threshold. Defect-removal ratio requires a reproducible defect measure. Collateral removal is evaluated outside the defect or target region using a declared mask. Core uniformity and transition error use the regions defined for h^*.

8.11 Statistical Protocol

The independent experimental unit, sample size, repetitions, randomization or counterbalancing, and exclusion criteria are [Planned/TBD: TBD]. Method order should control abrasive wear, temperature, and specimen variation. Paired analyses are preferred when methods can use matched defects or regions. Reports will include central tendency, dispersion, confidence intervals, the planned statistical test, effect sizes where meaningful, and multiple-comparison handling. Safety-stopped trials remain visible in success and intervention statistics.

8.12 Qualitative Analysis

Qualitative evidence includes synchronized policy trajectories, force traces, repetition behavior, actual execution paths, estimated and measured removal maps, core/transition profiles, residual action traces, and registered before/after surface observations. Failure cases, over-removal, contact loss, and policy saturation are shown alongside representative successes.

8.13 Current Result Status

The repository does not yet contain completed calibration, trial measurements, or statistical analysis for the redesigned scenario. No method is therefore described as outperforming another. All table cells are intentionally blank and will be populated only from experimental results with traceable raw data and metrology.

Chapter 9

Conclusions and Future Work

9.1 Conclusions

FAIRE separates localized surface repair into responsibilities that require different forms of reasoning. Force-aware expert imitation determines defect-specific selective treatment from visual context, robot state, and six-axis wrench history. It predicts short-horizon Cartesian pose and desired three-axis force trajectories, preserves the demonstrated tangential feed velocity, and executes desired contact force as a controller-level reference through admittance control.

Actual IL execution is then logged to estimate what the process physically produced. A calibrated Preston model and TIF convert contact pose, normal force, relative velocity, and dwell into a spatial removal map. A target profile regularizes the defect-containing core toward a uniform target depth and tapers removal smoothly across a transition band. The resulting residual-removal map drives structured model-based blending rather than uniform reprocessing.

During blending, admittance handles unknown surface height and bounded residual RL adapts roll, pitch, and local tangential feed velocity on unseen smooth surface geometries. This architecture is CAD-free in the sense that it avoids a complete CAD model or full reconstruction, while still using local path geometry and process models. The staged evaluation is defined, but quantitative conclusions remain [Planned/TBD: to be populated from experimental results].

9.2 Limitations

Removal accuracy depends on calibration of the Preston coefficient and on the validity of the TIF approximation. Force, orientation, pad wear, abrasive condition, and material response may change the influence function. Pose and metrology registration error further limit map accuracy and spatial resolution.

The system can remove material but cannot add it when IL over-removes. Conservative selection of h_c therefore depends on available coating or material thickness and uncertainty margins. Termination is not yet established as learned or autonomous. The supported geometry range is limited to local smooth surfaces with moderate continuous curvature; sharp edges, steps, grooves, discontinuities, and extreme curvature are outside the current scope.

Residual RL introduces sim-to-real and training-distribution limitations. Orientation and velocity residuals interact through contact and dwell, and their independent benefits cannot be assumed without ablation. Generalization claims are limited primarily to geometry while material, tool, abrasive, and process conditions remain fixed unless separate cross-condition studies are performed. Metrology resolution bounds the smallest defensible removal difference.

9.3 Future Work

Future work should add closed-loop post-pass visual or profilometric inspection, uncertainty-aware residual planning, online adaptation of K and the TIF, and calibrated models for multiple materials and coating systems. Safety-filtered residual RL and explicit uncertainty-aware termination could reduce the risk of irreversible over-removal.

The geometry scope may be expanded through richer local surface estimation and training on more complex smooth convex–concave combinations. More varied defect shapes, severity distributions, and removal difficulties should be evaluated after they can be produced and measured reliably. Iterative map updating and replanning should be implemented and validated before being promoted from a planned extension to a system contribution.

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Appendix A

Reproducibility and Completion Checklist

A.1 Demonstration Metadata

- episode, operator, specimen, defect, material, coating, tool, abrasive, and consumable identifiers;
- raw timestamps and rates for image, tracker, wrench, robot state, and command streams;
- raw and transformed pose/orientation, raw and processed six-axis wrench, contact labels, repetition annotations, and stopping events;
- episode acceptance, tracking loss, saturation, safety intervention, and rejection reasons;
- episode-level train, validation, and test assignments with leakage checks.

A.2 Calibration and Wrench Processing

- VR-to-robot, tracker-to-tool, sensor-to-tool, and TCP-to-contact-point transforms with conventions and validation residuals;
- sensor zeroing, wrench bias, gravity/tool-load compensation, spatial frame transformation, filter, delay, contact threshold, and sign convention;
- pad/tool geometry, spindle convention, tangential feed velocity computation, and true relative-velocity computation.

A.3 Removal Model and Metrology

- Preston coefficient K, units, calibration range, fitting method, and held-out validation;
- TIF calibration design, spatial parameterization, and normalization;
- TIF force/orientation dependence and uncertainty;
- metrology device, resolution, surface registration, sampling, preprocessing, repeatability, and reference surface;
- target-profile core, transition width, taper function, h_c selection, coating-thickness bound, and residual-map threshold;
- map grid, interpolation, uncertainty margin, and repeated-pass accumulation rule.

A.4 Planning, Control, and RL

- coverage path type, boundary rule, spacing, overlap, turn strategy, nominal force, nominal feed velocity, and pass-count rule;
- admittance frame, axes, discrete update, M_d, D_d, K_d, filter, saturation, reset, and contact-state logic;
- IL visual/state encoders, six-axis wrench-history encoder, history K, horizon H, outputs, loss, training settings, and stopping implementation;
- RL observation/state, action, reward, algorithm, network, training environment, privileged information, randomization, and sim-to-real procedure;
- roll, pitch, velocity, rate, force, moment, workspace, cumulative-removal, contact-loss, and emergency-stop bounds.

A.5 Experimental and Statistical Reporting

- robot, tool, sensor, camera, spindle, specimens, materials, abrasives, and environmental conditions;
- defect preparation, severity measurement, train/test surface geometries, and controls for material and tool condition;
- sample sizes, independent unit, randomization or counterbalancing, trial exclusions, failures, and safety-stopped trials;
- definition, units, instrument, aggregation, uncertainty, and analysis for every reported metric;
- statistical hypotheses, paired or unpaired design, confidence intervals, multiple-comparison handling, seeds, and raw-data provenance;
- every blank field in Tables 8.1, 8.2, 8.3, 8.4, 8.5, and 8.6.

A.6 Evidence Status

At the present draft stage, implementation-specific values and numerical outcomes are [Planned/TBD: to be determined from the final implementation] or [Planned/TBD: to be populated from experimental results]. Before submission, every placeholder must be replaced by a value traceable to code, configuration, calibration output, experiment log, or verified metrology record.

국문요약

FAIRE: CAD 형상 정보가 없는 로봇 표면 복원을 위한 힘 인지 모방학습 및 잔여 제거량 기반 블렌딩

국부 표면 결함의 복원에서는 손상되지 않은 주변 재료를 보호하면서 결함만 선택적으로 제거해야 한다. 일반적인 인식 및 경로 계획 기법은 결함의 위치를 찾고 기하학적 처리 경로를 생성할 수 있지만, 영상에서 확인되는 형상만으로는 결함의 깊이, 심각도, 제거 난이도에 따른 공정 강도, 반복 처리 위치, 접촉 거동, 종료 시점을 결정하기 어렵다. FAIRE는 이러한 불확실한 판단을 수행하기 위해 힘 인지 모방학습을 단순 경로 생성기가 아닌 전문가 복원 정책으로 사용한다. 정책은 영상, 로봇 상태, 최근 6축 렌치 이력을 관측하고 짧은 구간의 직교좌표 위치·자세와 3축 목표 접촉력 궤적을 예측한다. 이를 통해 작업자가 결함에 따라 선택한 처리 방식, 처리량, 국부 반복 및 종료 판단을 학습한다. 목표 접촉력은 액추에이터 명령이 아닌 제어기 수준의 기준이며, 선택적 제거 단계에서는 시연된 접선 이송 속도를 유지한다.

그러나 결함이 보이지 않게 제거되었다고 해서 표면 복원이 완성되는 것은 아니다. 결함 주변에 집중된 가공은 국부 함몰, 영역 내부의 불균일 제거, 방향 전환부의 중첩 제거, 미가공 표면과의 경계 단차 또는 외관 차이를 남길 수 있다. 이에 FAIRE는 모방학습 실행 중 실제 위치, 자세, 렌치 및 속도 이력을 기록하고, 보정된 Preston 계수와 공구 영향 함수(tool influence function)를 이용하여 실행 기반 재료 제거량 지도를 추정한다. 결함을 포함하는 중심 영역은 균일한 목표 깊이로 정규화하고, 바깥 전이 영역에서는 목표 제거량을 미가공 표면까지 부드럽게 감소시킨다. 목표 지도와 추정 지도 사이의 부족분은 잔여 제거량 지도로 정의된다.

모델 기반 계획기는 이 잔여 제거량으로부터 국부 2차원 블렌딩 경로와 공정 조건을 생성한다. 어드미턴스 제어는 미지의 표면 높이 변화를 보상하고, 제한된 잔차 강화학습은 블렌딩 단계에서만 롤·피치 자세와 접선 이송 속도를 조정하여 학습에 사용되지 않은 완만한 곡면에 대응한다. 평가는 전문가 모방학습, 힘 이력 표현, 제거 모델 보정, 블렌딩, 잔차 정책 및 전체 복원 과정의 단계별 실험으로 계획한다. 정량적 결과는 완료된 실험 자료를 바탕으로만

제시한다.

주요어: 로봇 표면 복원, 힘 인지 모방학습, 재료 제거량 추정, 모델 기반 블렌딩, 어드미턴스 제어, 잔차 강화학습, CAD 비의존 표면 가공